

OlmoEarth: Stable Latent Image Modeling for Multimodal Earth Observation

Henry Herzog^{1*} Favyen Bastani^{1*} Yawen Zhang^{1*} Gabriel Tseng^{1*} Joseph Redmon^{1*}
 Hadrien Sablon¹ Ryan Park¹ Jacob Morrison^{1,2} Alexandra Buraczynski¹ Karen Farley¹
 Joshua Hansen¹ Andrew Howe¹ Patrick Alan Johnson¹ Mark Otterlee¹ Ted Schmitt¹
 Hunter Pitelka¹ Stephen Daspit¹ Rachel Ratner¹ Christopher Wilhelm¹ Sebastian Wood¹
 Mike Jacobi¹ Hannah Kerner³ Evan Shelhamer⁴ Ali Farhadi^{1,2} Ranjay Krishna^{1,2} Patrick Beukema^{1*}

¹Allen Institute for AI ²University of Washington ³Arizona State University ⁴University of British Columbia

*Equal contribution from modeling team. Authors listed in reverse alphabetical order by last letter of first name.

olmoearth@allenai.org

Abstract

Earth observation data presents a unique challenge: it is spatial like images, sequential like video or text, and highly multimodal. We present OlmoEarth: a spatio-temporal, multimodal foundation model that employs a novel self-supervised learning formulation, masking strategy, and loss, which are all designed for Earth observation. OlmoEarth achieves state-of-the-art performance compared to 12 other foundation models across a variety of research benchmarks and real-world tasks from external partners. When evaluating task-agnostic embeddings OlmoEarth achieves the best performance on 15 out of 24 tasks, and with task-specific full fine-tuning it is the best on 19 of 29 tasks. We deploy OlmoEarth as the backbone of an end-to-end platform for data collection, labeling, training, and inference of Earth observation models. The OlmoEarth Platform puts frontier foundation models and powerful data management tools into the hands of non-profits and NGOs working to solve the world's biggest problems. OlmoEarth source code, training data, and pre-trained weights are available at https://github.com/allenai/olmoearth_pretrain.

1. Introduction

Earth observation foundation models show promising results in research settings [2, 15, 16, 27, 53, 57, 58]. However, adoption for real-world tasks lags behind, especially in the non-profit sector. Foundation models are large, complex to train, and expensive to deploy.

To help enable non-profit, humanitarian, and environmental organizations to use these powerful tools, we train OlmoEarth, a new family of models, using a novel, stable

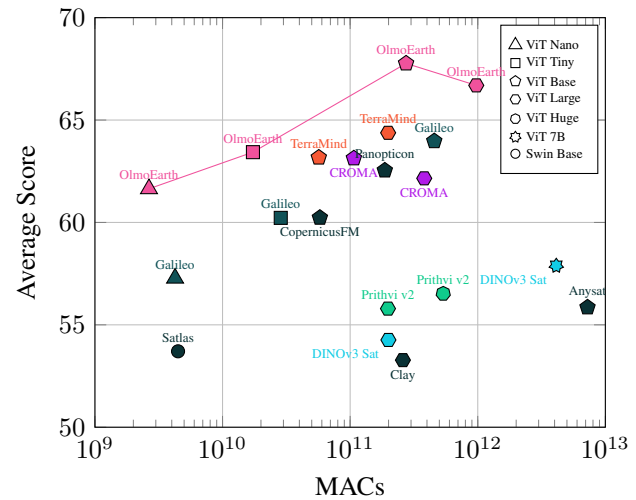


Figure 1. OlmoEarth defines a Pareto optimum of performance vs. computational efficiency averaged across 13 embedding tasks (measured by kNN and linear probing)¹. The chart shows average multiply-accumulate operations to encode one example across all tasks (input size varies by task). See Table 2 for full results.

training regime. We comprehensively evaluate OlmoEarth against 12 other foundation models on research benchmarks and real-world tasks from partner organizations. Finally, we deploy these models in an open, end-to-end platform bringing frontier models directly to organizations who need it the most.

When attempting to replicate existing work, we frequently saw training instability, representation collapse, and models underperforming their stated potential. Our approach, **Latent MIM Lite**, strikes a middle ground between masked autoencoders (MAE), which are stable but limited in feature quality, and latent approaches like I-JEPA and La-

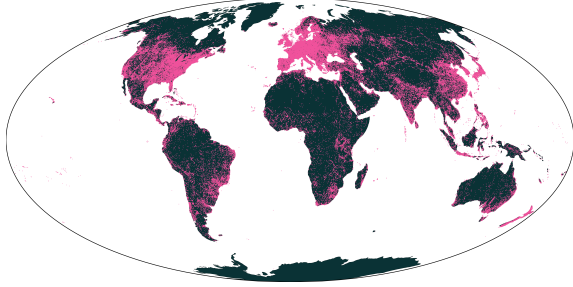


Figure 2. Global distribution of OlmoEarth pretraining data. We sample 285,288 locations based on OpenStreetMap categories.

tent MIM, which produce better features but are unstable [1, 36, 59]. Latent MIM Lite replaces the target encoder with a frozen, randomly initialized linear projection, stabilizing training while maintaining the representative power of latent modeling. We further introduce a modality-aware masking strategy and a bandset-contrastive loss tailored to the multimodal nature of Earth observation data (Section 2).

We evaluate OlmoEarth against 12 other foundation models on 18 research benchmarks and 19 datasets from 7 partner organizations using standard transfer learning techniques (kNN and linear probing) and full, end-to-end fine-tuning [16, 41, 53]. OlmoEarth achieves the best performance in 15 of 24 embedding tasks and 19 of 29 fine-tuning tasks (Figure 1).

We deploy OlmoEarth as part of the OlmoEarth Platform, an end-to-end solution for organizations harnessing Earth observation data for the public good. Our partners are already using the platform for mangrove conservation, ecosystem mapping, and food security. The OlmoEarth Platform solves the last-mile problem of putting frontier research into the hands of people who can use it to do the most good.

2. OlmoEarth

OlmoEarth is a Vision Transformer (ViT) based encoder-decoder style architecture. It processes a multimodal image timeseries of aligned satellite images and derived maps. A FlexiViT-style projection layer [5] converts the input data from pixels to tokens with a variable patch size. Positional, temporal, and modality encodings add additional context to the tokens. During training, some portion of the input tokens are masked. The encoder transformer layers attend across space, time, and between modalities to produce embeddings for the input tokens. The decoder predicts representations for the masked input tokens.

¹Average over all tasks every model can perform, specifically the Sentinel-2 versions of: m-bigearthnet, m-so2sat, m-brick-kiln, m-eurosat, BreizhCrops, CropHarvest-Togo, CropHarvest-PRC, m-cashewplant, m-SA-crop-type, PASTIS, MADOS, AWF, Nandi.

2.1. Data

OlmoEarth is designed to flexibly handle input Earth observation data across a range of spatial and temporal resolutions. During our pretraining experiments we train on three satellite modalities and six derived maps:

Observations	Maps	
Sentinel-1	WorldCereal [56]	OpenStreetMap [40]
Sentinel-2	WorldCover [64]	Cropland Data Layer [54]
Landsat-8	SRTM [38]	Canopy Height Map [50]

Our pretraining dataset contains 285,288 samples from around the world. Each sample covers a $2.56\text{km} \times 2.56\text{km}$ spatial region and a one-year time range. For multi-temporal modalities, we use up to 12 timesteps sampled monthly over the course of the year, although many samples contain only a subset of the timesteps and modalities.

For the above modalities we resample the data to be uniformly 10 meters per pixel. We experimented with adding NAIP data at 2.5 meters per pixel [55] and ERA5 data at 160 meters per pixel [25] but found no significant improvement on our evaluations.

We further subdivide Landsat and Sentinel-2 into **bandsets** based on the original resolution of their bands, grouping bands captured at the same resolution together. Landsat consists of 2 bandsets while Sentinel-2 consists of 3 bandsets. For the precise split see the OlmoEarth source code. The embedding layer has per-bandset weights so one token is produced per patch $\times$ timestep $\times$ bandset.

The locations of samples are chosen based on OpenStreetMap features. We select 120 categories of map features in OpenStreetMap, ranging from roads to geothermal power plants, and enumerate all $2.56\text{km} \times 2.56\text{km}$ tiles containing each category. We then randomly sample up to 10,000 tiles per category to derive the 285,288 samples (many categories appear in fewer than 10,000 tiles). The one-year time range of each sample is sampled uniformly between January 2016 and December 2024.

2.2. Architecture

Similar to many Earth observation models, OlmoEarth is a transformer-based encoder-decoder style architecture. Inspired by Galileo, we use a flexible patch-embedding layer [5, 53]. OlmoEarth uses a flexible embedding strategy similar to FlexiViT. Instead of resizing the filter weights, it resizes the input image to simulate smaller or larger patch sizes. This optimization avoids a GPU sync in our implementation. We verified that this produces near-identical performance to standard FlexiViT resizing in both loss and downstream tasks.

Once the input is in token space, OlmoEarth adds in a 2D sincos positional embedding, a sinusoidal temporal embedding, and a learnable modality embedding to each token.

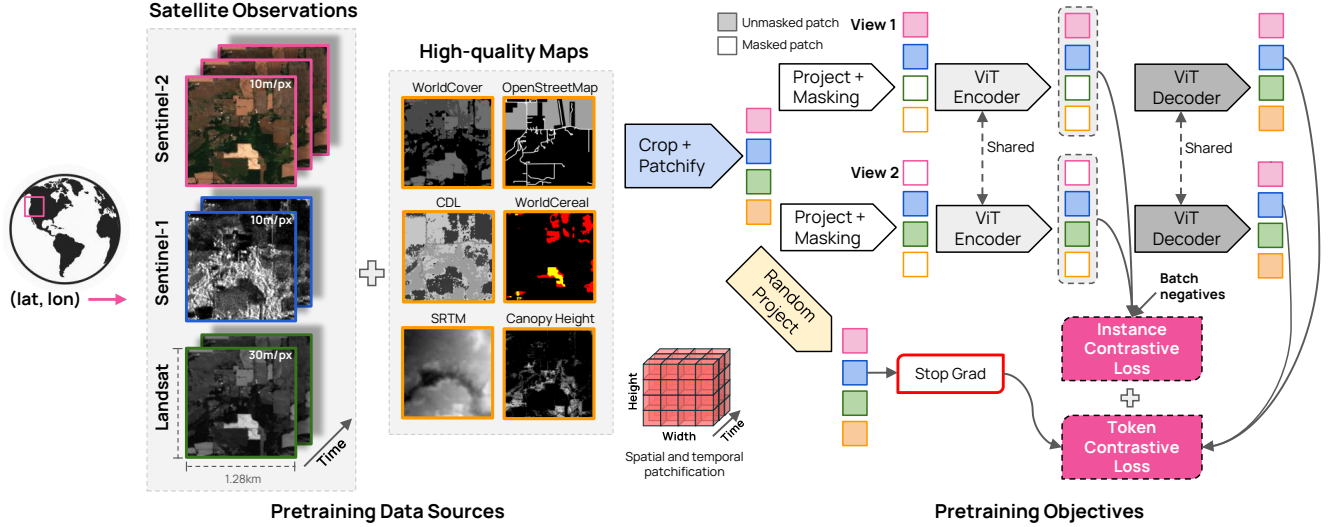


Figure 3. We train OlmoEarth with a combination of satellite observations and high-quality maps. After tokenizing these inputs, we: (1) apply a modality-aware masking strategy to define which tokens are inputs vs. targets, (2) pass the target tokens through fixed random projections to construct targets, (3) pass the input tokens through our learned encoders, and then (4) through a decoder which predicts the target tokens and (5) apply a modality-aware patch discrimination loss between the predicted and target tokens. Steps 1-5 are applied twice on the same data to then (6) apply an instance contrastive loss over the aggregated tokens per instance.

During training, some tokens are masked out of the input, otherwise all tokens are passed to the encoder transformer which performs full self-attention across space, time, and between modalities.

Architecture	Depth	Dim	Heads	Parameters
ViT Nano	4	128	8	1.4M
ViT Tiny	12	192	3	6.2M
ViT Base	12	768	12	90M
ViT Large	24	1024	16	300M

Table 1. ViT encoder architectures and number of parameters for the four OlmoEarth model sizes.

We train four different encoder sizes based on standard Vision Transformer sizes, see Table 1. For each model size, the decoder has the same feature dimension and number of heads but only a depth of 4. We design a smaller decoder so that the encoder does the majority of the modeling.

During training the decoder represents the masked portions of the input with a learned $\langle \text{MASK} \rangle$ token added to the appropriate positional, temporal, and modality embeddings. The decoder cross-attends to these tokens with the visible tokens from the encoder. It then predicts the latents for the masked tokens.

2.3. Masking

OlmoEarth uses a modality-aware masking strategy. For every example the masking strategy selects some bandsets to be encoded and also some to be decoded, non-exclusively.

Thus every bandset falls into one of four categories:

- **Not selected:** Ignored for this example.
- **Encode only:** Randomly masked, input to encoder.
- **Decode only:** Used as target for decoder.
- **Encode and decode:** Randomly masked, input to encoder, masked tokens used as targets for decoder.

This masking strategy re-frames the problem slightly from reconstructing data that has been partially masked to reconstructing missing bandsets from partial views of other bandsets. When all bandsets are encoded and decoded we find the task is too easy. Masked tokens in a bandset will likely have other tokens in the same bandset that are highly correlated with them that are visible in the input, tokens nearby spatially or temporally. Training in this easier paradigm requires using very high masking ratios (i.e. masking out 90% of the input) to get decent results. Masking some bandsets entirely makes the problem harder and allows more balanced masking ratios.

OlmoEarth trains on both observations and maps but at inference time we only use observations. Maps can change over time—indeed downstream tasks are often detecting this kind of change—so we only rely on observations for inference. Thus, during training our masking strategy never encodes map data, it only ever decodes it. While observations can fall into any of the above four categories, maps will only be “decode only” or “not selected”. We make a similar design choice to AlphaEarth, where certain modalities are used only as training targets and are not processed by the online encoder.[9] During training the frozen target encoder

projects maps into token space but maps are not processed by the online encoder; they are only used as targets for reconstruction.

2.4. Latent MIM Lite

During training OlmoEarth predicts reconstructions of the masked input in latent space. The target encoder is a copy of the embedding layer from our online encoder that is randomly initialized and frozen during training. We use this frozen projection layer for each modality to project masked patches in the input into token space. Thus, OlmoEarth performs Latent Masked Image Modeling of Linear, Invariant Token Embeddings (Latent MIM Lite).

Randomly projecting raw input data extracts valuable features both from a theoretical and practical standpoint [6, 7, 45]. As such, random projections provide a non-trivial latent space that is relevant for predictions from our input data. However, because we use a fixed target encoder we avoid the representation collapse common in Latent MIM-style training. While it's possible this approach is too simplistic in more diverse domains like natural image processing, empirical results show a clear benefit in our domain of Earth observation data.

Latent MIM Lite allows us to unify supervised and self-supervised training under the same architecture. We project each modality, whether observations or maps, through a frozen random projection into token space. Loss is calculated the same for both types of modalities. We do not need to add on specific predictor heads for supervised data or adjust our training strategy or loss. In our ablations we see this approach gives strong results in a purely self-supervised setting and also benefits from additional supervised data.

Other models like Galileo and TerraMind train on both supervised and unsupervised data, but they treat supervised maps as a valid input to the model [27, 53]. This means their encoders must learn to model these map modalities as input and during training may use map modalities to predict observations or other map modalities. While this also unifies supervised and semi-supervised training, we theorize that our approach simplifies learning for the encoder while maintaining the benefits of training with supervised data. In our evaluations we see improved performance over these models on most tasks.

2.4.1. Modality Patch Discrimination

Masked image modeling in pixel space typically uses a reconstruction loss like Smooth L1 [18]. Latent MIM proposes using a contrastive loss (Patch Discrimination) instead of reconstruction loss to incentivize diversity in the latent space predictions. Patch discrimination loss frames token reconstruction as a classification task where we want the predicted token for a patch to be similar to the target token but dissimilar from other ground truth tokens for other

patches. Patch discrimination uses cosine similarity to measure token similarity and cross entropy loss to contrast between positive and negative matches.

Typical patch discrimination contrasts a predicted token with all target tokens in the input. For image modeling, the target tokens from an image are encodings of different parts of the image so they are from the same distribution, making the contrastive task challenging. In OlmoEarth, different target tokens can come from different modalities or different time steps as well as different spatial locations.

Tokens from different modalities have very different distributions so distinguishing between them is easy. Yet there are so many tokens from other modalities that a significant amount of the loss comes from these “easy” negatives. We find eliminating easy negatives and only contrasting tokens with targets from the same modality gives a substantial performance increase.

2.4.2. Instance Contrastive Loss

The patch discrimination loss operates on the local representations generated by the encoder and decoder but many tasks (like classification) require a global understanding of the input. Some foundation models use a single `<CLASS>` token to represent this global information. Instead we opt to pool information globally over all modalities, timesteps, and locations for an input. To generate a global representation for an input we run the OlmoEarth encoder and average pool the output tokens.

Tokens encoded from the same modality share semantics but tokens from different modalities may look very different from each other. We want to be able to average tokens from all modalities together and get a sensible global representation of an input. Thus, we use a contrastive loss on the pooled representation from the encoder to encourage tokens to exist in a common representation space and behave well when pooled.

We want both positive and negative samples for our contrastive loss so we take an approach similar to SimCLR [10] and encode two versions of the same input, contrasting these two versions as positive examples with the rest of the batch as negative examples. However, instead of using different data augmentation to generate the two samples we use different random masking.

We run random masking twice, then encode both batches with our encoder, pool the resulting tokens, and apply contrastive loss to the pooled representations. We run the decoder twice, decoding masked portions for both images and calculate the modality patch discrimination loss. A scalar multiple controls the contribution of instance contrastive loss to modality patch discrimination loss. For experiments in this paper we scale the instance contrastive loss by 0.1.

3. Experiments

We extensively evaluate OlmoEarth on both standard research benchmarks and real-world downstream tasks from partner organizations. Following standard practice in remote sensing foundation models, we evaluate both kNN/linear probe performance with a frozen encoder and full fine-tuning performance [16, 41, 53].

To get as comprehensive an evaluation as possible we import other top performing foundation models into our evaluation framework and evaluate them as well so they are directly comparable [2, 4, 11, 15, 16, 27, 46, 49, 52, 53, 57, 58]. We use the same training recipes for each foundation model but sweep a variety of hyperparameters to find the best performance for each model on each task. We leave evaluations blank for models that do not support particular modalities. We also do not fine-tune some large models on partner tasks due to compute and time limitations.

3.1. Pretraining

We pretrain OlmoEarth on our pretraining dataset described in 2.1 using Latent MIM Lite. We use AdamW optimization with a base learning rate of 1×10^{-4} , weight decay of 0.02, batch size of 512, linear learning rate warm-up of 8000 steps, cosine annealing of learning rate by 0.1 over a total of 667, 200 steps. Due to memory constraints we use a micro-batch size of 32 so the pooled contrastive loss is only applied over these 32 examples, not the full batch of 512.

During training OlmoEarth uses a random effective patch size in the range $\{1 \dots 8\}$ and takes a random square crop from the input with side length in tokens in the range $\{1 \dots 12\}$. Thus, along the spatial dimension the smallest input is a 1×1 pixel region in the input with a patch size of 1, and the largest input is 96×96 pixel region in the input with a patch size of 8. Along the temporal dimension, our model processes between 3 and 12 timesteps. During training our model processes around 100 billion tokens.

3.2. Research Benchmarks and Partner Tasks

We evaluate on a variety of common research benchmarks for classification and segmentation across single and multiple sensor modalities. Our evaluations include all seven Sentinel-2 and Landsat benchmarks from GEO-Bench [32]: m-bigeearthnet, m-so2sat, m-brick-kiln, m-forestnet, m-eurosat, m-cashewplant, and m-SA-crop-type. We also evaluate on the classification benchmarks BreizhCrops [44] and CropHarvest [51] and the segmentation benchmarks PASTIS [17], MADOS [31], and SenIFloods11 [8].

While developing OlmoEarth, we partnered with several organizations who are already using or want to use remote sensing data for environmental, climate, or research tasks. These organizations provided labeled data across a variety of domains for our evaluations, offering critical insights into how models perform on real-world tasks. For example,

we partnered with the African Wildlife Foundation (AWF) to map land use and land cover in southern Kenya. We pair these tasks with different combinations of Sentinel-1, Sentinel-2, and Landsat observations.

3.3. kNN and Linear Probing

For evaluations without fine-tuning we extract embeddings from the train, validation, and test set and apply either a kNN model for single time step classification or a linear probe model for segmentation and multi-temporal classification. For OlmoEarth we use a patch size of 4 except we sweep patch size for applicable models on m-Cashew Plant (See discussion in Appendix). For external models we use recommended settings for patch size and resize input data to that model’s pretraining size following [13]. For models that do not support time series data we input each time step separately. To aggregate embeddings on time-series input, we average pool across timesteps. We sweep pooling method for the resulting embeddings across time (mean vs max). We also sweep normalization statistics (computed during pretraining vs. on the evaluation set).

We run kNN with $k = 20$ using cosine similarity, and follow standard evaluation practices [22, 53]. For models that output a `<CLASS>` embedding token we use that as the embedding for the whole image, otherwise we average across resulting tokens.

We run linear probing on the output embeddings, training for 50 epochs. We sweep across a variety of learning rates for each model $\{1 \times 10^{-4}, 5 \times 10^{-4}, 1 \times 10^{-3}, 5 \times 10^{-3}, 1 \times 10^{-2}, 5 \times 10^{-2}, 1 \times 10^{-1}, 5 \times 10^{-1}\}$ and report the test results for the highest validation set performance.

3.4. Fine-Tuning

For fine-tuning evaluations, for each model, we take the encoder and add a decoder that makes classification, regression, semantic segmentation, or object detection predictions. Our fine-tuning recipe freezes encoder parameters for 20% of the epochs, only training the added decoder layers, and then unfreezes and fine-tunes the full model for the remaining epochs. We use AdamW optimization with a plateau scheduler that reduces the learning rate by a factor of 0.2 after 2 epochs without improvement on the validation set and a 10 epoch cooldown after reduction.

For fine-tuning on research benchmarks, the decoder is a single-layer linear probe; for classification tasks, it makes a prediction using embeddings pooled over the image, and for segmentation tasks, it makes a prediction using embeddings pooled temporally (when applicable) at each spatial patch. We sweep learning rates for each model over $\{1 \times 10^{-4}, 5 \times 10^{-4}, 1 \times 10^{-3}\}$.

For fine-tuning on partner tasks, the decoder is:

- **Classification:** 3-layer MLP.

Model	Modalities Time series Method Metric	m-bispectral		m-bisect		m-bx4x4h		m-forest		m-moment		m-bboxCrop		CropHarvest-PRC		CropHarvest-PRC		CropHarvest-PRC		CropHarvest-Togo		CropHarvest-Togo		m-scale-plant		m-s4-crop-type		PASTIS		PASTIS		MADOS		SciFlux11		AWF		AWF		AWF		Nandi		Nandi		Nandi	
		S2	S2	S2	L8	S2	S2	S2	S2	S1	S2	S1,S2	S1	S2	S1,S2	S1	S2	S1,S2	S1	S2	S1,S2	S1	S2	S1	S2	S2	S2	S1	S2	S2	S2	S1	L8	S1	S2	L8	S1	S2	L8	S1	S2						
		✗	✗	✗	✗	✗	✗	✗	✗	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✗	✓	✓	✗	✓	✓	✗	✗	✓	✓	✗	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓			
		kNN	kNN	kNN	kNN	kNN	kNN	LP	LP	LP	LP	LP	LP	LP	LP	LP	LP	LP	LP	LP	LP	LP	LP	LP	LP	LP	LP	LP	LP	LP	LP	LP	Acc.	Acc.	Acc.	Acc.	Acc.	Acc.	Acc.	Acc.	Acc.	Acc.	Acc.	Acc.			
		μF1	Acc.	Acc.	Acc.	Acc.	Acc.	Acc.	Acc.	Acc.	Acc.	Acc.	Acc.	Acc.	Acc.	Acc.	Acc.	Acc.	Acc.	Acc.	Acc.	mIOU	mIOU	mIOU	mIOU	mIOU	mIOU	mIOU	mIOU	mIOU	mIOU	mIOU	Acc.	Acc.	Acc.	Acc.	Acc.	Acc.	Acc.	Acc.	Acc.	Acc.	Acc.	Acc.			
AnySat	ViT Base	54.5	36.5	84.5	34.0	80.4	62.7	57.0	74.1	72.3	69.6	68.4	74.5	24.6	27.2	24.2	41.9	41.3	77.8	-	60.0	60.0	64.0	57.2	20.5	55.8	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-		
	Clay	48.8	38.7	90.8	40.3	86.3	57.0	66.7	66.5	63.4	78.4	67.3	78.4	67.3	30.8	23.1	19.9	22.6	47.5	78.9	59.5	61.5	56.5	52.4	23.4	57.0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-			
	CopernicusFM	ViT Base	64.6	50.3	85.9	-	84.7	65.5	55.1	72.8	74.4	77.5	75.8	70.6	32.2	28.4	15.9	32.1	63.9	77.6	-	59.0	67.0	-	24.4	59.8	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-		
	CROMA	ViT Base	61.3	51.3	92.0	-	84.6	69.0	56.8	74.3	75.1	76.8	80.7	76.5	24.9	30.3	26.3	44.7	60.4	78.8	-	67.5	79.0	-	24.6	68.2	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-		
CROMA	ViT Large	59.2	48.2	91.7	-	85.8	68.2	56.2	72.7	71.0	79.4	76.5	81.0	27.0	30.4	25.9	42.7	66.4	78.8	-	62.5	71.0	-	26.1	68.2	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-		
	DINOv3	ViT Base	51.0	47.1	91.3	43.3	86.6	31.3	-	64.5	-	-	67.0	-	23.5	26.7	-	18.1	53.5	-	48.5	-	61.0	42.5	-	54.1	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-		
	DINOv3	ViT Large	55.8	45.3	89.9	46.0	84.0	31.3	-	66.1	-	-	68.3	-	24.5	26.1	-	17.4	52.4	-	43.5	-	60.5	39.9	-	49.8	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-		
	DINOv3	ViT Huge+	57.0	45.7	88.2	46.9	86.1	31.3	-	68.7	-	-	68.3	-	25.1	26.7	-	17.4	48.1	-	47.5	-	54.0	43.5	-	55.4	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-		
DINOv3	ViT 7B	60.8	46.6	91.3	48.0	85.2	31.3	-	68.7	-	-	70.9	-	34.3	27.9	-	21.1	52.2	-	47.5	-	57.0	44.7	-	56.7	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-		
	DINOv3 Sat	ViT Large	60.2	44.0	91.4	44.2	89.2	31.3	-	70.1	-	-	68.6	-	32.4	28.5	-	22.5	57.5	-	42.5	-	69.5	35.5	-	48.4	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-		
	DINOv3 Sat	ViT 7B	61.6	50.1	91.4	47.0	91.3	31.3	-	72.2	-	-	71.9	-	54.1	31.7	-	26.3	59.7	-	49.5	-	68.5	31.8	-	42.5	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-		
	Galileo	ViT Nano	55.0	53.7	90.9	-	89.4	66.3	60.9	74.4	72.4	75.2	70.3	78.1	21.2	19.5	19.2	19.1	53.1	78.6	-	65.5	67.5	-	24.4	64.2	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-		
Galileo	ViT Tiny	55.8	53.1	87.5	-	89.1	66.7	55.7	80.3	79.3	69.9	78.8	77.1	23.6	21.5	23.4	27.7	61.1	78.6	-	65.5	71.0	-	25.0	66.7	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-		
	Galileo	ViT Base	58.3	55.7	91.1	-	92.8	69.7	60.9	81.9	79.3	67.3	80.1	77.5	28.9	25.3	28.0	39.6	68.4	79.4	-	66.5	72.5	-	26.8	67.3	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-		
	Panopticon	ViT Base	64.9	60.5	92.9	52.3	95.2	57.7	55.9	75.9	75.6	72.2	72.9	76.5	32.7	27.3	23.7	30.2	66.1	78.0	-	66.0	65.0	71.5	60.4	22.9	65.2	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-		
	Presto	ViT Nano	-	-	-	-	-	60.9	59.3	74.3	76.6	78.4	81.4	72.5	-	-	16.3	28.2	-	-	-	-	60.5	53.5	-	25.2	57.6	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-		
Prithvi v2	ViT Large	51.6	34.7	89.7	37.9	82.2	66.2	-	67.6	-	-	71.9	-	46.8	24.7	-	37.2	50.9	-	57.0	-	49.0	57.7	-	52.8	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-		
	Prithvi v2	ViT Huge	51.0	34.0	90.1	41.4	81.2	66.1	-	69.5	-	-	69.9	-	45.8	26.6	-	37.5	52.2	-	59.0	-	55.0	57.9	-	55.9	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-			
	Satlas	Swin Base	52.3	44.5	83.0	36.9	82.6	66.6	57.4	71.3	-	76.8	75.8	-	30.6	24.0	10.5	14.4	30.2	72.9	57.5	52.0	62.0	61.5	25.1	60.2	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-			
	TerraMind	ViT Base	63.9	46.7	91.9	-	85.6	66.4	57.0	75.1	75.6	74.2	74.8	77.5	46.0	30.4	22.7	40.9	66.0	78.7	-	66.0	69.5	-	24.7	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-			
TerraMind	ViT Large	63.9	47.4	92.2	-	90.0	68.2	56.2	74.7	72.0	75.2	74.8	77.5	50.4	31.2	22.3	41.3	67.5	78.4	-	62.0	67.0	-	23.3	65.3	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-		
	TESSERA	-	-	-	-	-	-	-	-	-	72.2	-	81.0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-			
	OlmoEarth	ViT Nano	59.5	54.3	96.2	38.8	89.9	64.1	58.0	79.5	74.3	73.5	81.7	83.7	25.5	23.6	18.1	35.0	55.2	78.2	73.0	61.5	69.5	57.7	24.8	67.4	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-		
	OlmoEarth	ViT Tiny	59.4	61.8	92.0	40.5	91.6	64.0	58.3	78.6	80.3	75.8	85.6	82.4	24.7	23.2	21.4	40.1	58.6	78.5	75.5	64.0	76.0	60.7	24.7	69.0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-		
OlmoEarth	ViT Base	62.4	67.7	93.3	41.9	94.7	70.9	56.8	73.4	75.4	80.1	87.3	82.0	32.3	28.9	29.7	50.6	67.2	79.2	77.0	68.5	77.5	67.9	26.5	74.7	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	
	OlmoEarth	ViT Large	62.0	68.2	93.4	41.6	96.3	70.7	56.6	74.1	75.1	67.6	78.1	79.7	30.9	28.5	30.6	51.8	66.4	79.8	76.0	66.5	73.0	66.4	26.2	73.3	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-		

Table 2. kNN/Linear probe results on research benchmarks and real-world tasks from our partners. We run kNN on single time-step classification tasks and linear probing on all other tasks. We sweep across data normalization strategies, feature pooling, and learning rate (for linear probing) and report the test set result for the best validation set performance. Not all models can run on all tasks due to incompatible input modalities. OlmoEarth has consistently strong performance and is the best on 15 out of 24 tasks.

Model	Modalities Time series Metric	eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer										eigenformer									
-------	-------------------------------------	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--	-------------	--	--	--	--	--	--	--	--	--

Table 3. Fine-tuning results on research benchmarks (left) and partner tasks (right). We train all models with the same recipe and report test set results for the model checkpoint with the best validation set performance. Some models are only compatible with a subset of tasks. Due to resource constraints, we do not fine-tune large models on all tasks. OlmoEarth is best on 19 out of 29 tasks.

- **Segmentation:** Transposed convolutional layers, or U-Net decoder for multi-scale encoders [43].
- **Object Detection:** Faster R-CNN head, with an FPN for multi-scale encoders [34, 42].

We use a learning rate of 10^{-4} for all tasks, except Nandi, for which some models exhibit unstable learning and we sweep over $\{10^{-4}, 10^{-5}\}$.

3.5. Results

For kNN/LP evaluations, OlmoEarth is the best performing on 11 of 18 research benchmarks and 4 of 6 partner tasks. For fine-tuning evaluations, OlmoEarth is the best performing on 5 of 10 research tasks and 14 of 19 partner tasks. OlmoEarth gets consistently high performance except in a couple instances.

OlmoEarth Large is the best-performing large model in the literature in aggregate, yet it does not always outperform

	m-so2sat	m-eurosat	PASTIS
Full Latent MIM*	32.2	68.4	7.9
Latent MIM Lite	42.2	87.2	35.2
+ Modality Masking	53.6	90.2	46.6
+ Modality Patch Disc	55.3	91.5	48.1
+ Contrastive Loss	56.8	92.3	49.0
+ Maps	62.4	92.9	50.7
Encode Maps	54.7	91.8	45.9

Table 4. Development path of the OlmoEarth base model showing effect of adding our various contributions starting from a Latent MIM approach. *Full Latent MIM collapsed during training.

OlmoEarth Base, particularly for embedding-based pixel time series tasks where it is significantly worse. This scaling challenge is not unique to OlmoEarth: CROMA Large performs worse than CROMA Base in aggregate on embedding tasks; TerraMind Large underperforms its Base on 9 of 20 embedding and 9 of 18 fine-tuning tasks; and DINOv3 Large is worse than its Base on 10 of 16 tasks. While all model families show good Nano→Tiny→Base scaling, the Base→Large transition involves tradeoffs across all model families, suggesting fundamental challenges in scaling Earth observation models [61].

Other notable models include Panopticon for strong performance on embedding tasks and Terramind on fine-tuning tasks. DINOv3 shows good results for tasks that mainly require visual information but lags behind specialized models on tasks where temporal understanding is critical. Galileo shows strong performance on many benchmarks, especially agriculture-related tasks.

3.6. Ablations

We based OlmoEarth on Latent MIM self-supervised training and iterated on various modifications, keeping the best. Table 4 shows our development process, starting from standard Latent MIM, random masking, patch discrimination loss only, and no maps data. Models in the table are trained according to training recipe in Subsection 3.1 but only for 140,000 steps. Results are shown for kNN and LP on the validation set of three benchmarks. During development we ran a subset of our evaluations in our “in-loop evals” but saw that improvements on a representative subset carried over to the full evaluation.

We see the Latent MIM model gets poor performance due to representation collapse. Switching to Latent MIM Lite substantially boosts performance. Further modifications show increased performance for all tasks. We also include an “Encode Maps” ablation that processes maps through the online encoder rather than using them only as decode targets; the resulting drop in performance supports our decode-only design for maps. We conduct additional ablations in Appendix E.

3.7. Environmental Impact

Following recent work on environmental impact analysis of language modeling [19, 37, 39] we estimate total energy use, carbon emissions, and water consumption from training OlmoEarth in Table 5. Similar to other environmental impact estimates this should be viewed as a lower bound as it does not account for hardware manufacturing, transportation, etc.

Model	Stage	Hardware	GPU Hrs	Energy (kWh)	Carbon (tCO ₂ eq)	Water (kL)
OlmoEarth Nano	Pretraining	H100	1,149	195	0.08	0.30
OlmoEarth Tiny	Pretraining	H100	1,149	205	0.08	0.32
OlmoEarth Base	Pretraining	H100	2,989	803	0.32	1.24
OlmoEarth Large	Pretraining	B200	5,240	1,933	0.77	2.99
OlmoEarth Nano	Fine-tuning	–	647	186	0.07	0.29
OlmoEarth Tiny	Fine-tuning	–	723	261	0.10	0.40
OlmoEarth Base	Fine-tuning	–	1,224	685	0.27	1.06
OlmoEarth Large	Fine-tuning	–	58	39	0.02	0.06
Total	Overall	–	13,179	4,307	1.72	6.67

Table 5. Approximate environmental impact of pretraining and fine-tuning OlmoEarth. Metrics for fine-tuning OlmoEarth Nano, Tiny, and Base include research benchmarks and partner tasks. Metrics for fine-tuning OlmoEarth Large only include research benchmarks.

We train all of our models in a single data center on NVIDIA H100 and B200 GPUs. We calculate the total GPU power required for a training run by tracking actual GPU power utilization every ~25ms to calculate a weighted average of power consumption throughout training. We then multiply this by the power usage efficiency (PUE) factor for our data center, according to our provider, and then we multiply this final GPU power usage amount by either the carbon intensity of the grid or the water usage efficiency factor of the data center to calculate total carbon emissions and water consumption, respectively.

The total energy usage during training (4,307 kWh) could power the average U.S. household for 5 months. The total carbon emissions are equivalent to an economy ticket on a flight from Seattle to Portugal.

4. Related Work

Pretraining for remote sensing models initially focused on contrastive approaches [3, 28, 35]. Recently masked modeling has taken over as the dominant paradigm, similar to language and vision [14, 23]. Early approaches to remote sensing pretraining directly reconstructed the masked pixel values [12, 41, 52]. Following research in natural imagery [1, 46, 59], remote sensing focuses more on reconstruction in latent space. Latent approaches work well [2, 53, 57] but have documented instabilities [1, 36].

TerraMind avoids instability by using a frozen tokenizer during pretraining. For image modalities they train a quantized autoencoder and use the encoder as their frozen tokenizer during multimodal masked modeling.

Sensor Generalization. Several recent models aim to generalize across arbitrary sensors. DOFA [62] uses dynamic adaptation modules to handle any optical sensor at inference time. Panopticon [57] trains a single model that accepts any optical sensor through a flexible input projection. RingMo [48] demonstrates large-scale masked image modeling for remote sensing, and SkySense [21] combines multiple modalities into a unified multi-modal foundation model. OlmoEarth focuses on achieving the best performance on widely-used sensors (Sentinel-1, Sentinel-2, Landsat) that appear most frequently in research evaluations and partner tasks, rather than targeting arbitrary sensor compatibility.

Precomputed embeddings offer an alternative approach for accessibility [9, 15] but still require expertise to retrieve and use. Best results may still require training a decoder on top of the embeddings. Precomputed embeddings also limit flexibility; both AEF and TESSERA generate annualized embeddings making real-time or sub-annual predictions impossible. OlmoEarth embeddings match or outperform AEF embeddings on partner tasks, and full fine-tuning enables even better results (Appendix Table 9).

5. Discussion

We want OlmoEarth to have a positive impact on the world. Toward that end, we release it as part of the OlmoEarth Platform, an end-to-end, open solution for Earth observation tasks. OlmoEarth Platform enables partner organizations to use the latest, best foundation models in their work on the environment, conservation, food security, and more. Organizations like Global Mangrove Watch, Global Ecosystem Atlas, and the International Food Policy Research Institute are using OlmoEarth Platform for data curation and labeling, model fine-tuning, and inference.

5.1. Case Studies

Global Mangrove Watch maps and tracks the extent and health of coastal mangrove forests. Mangrove forests sequester carbon, protect the coastline from erosion, and provide a habitat for little fishies. GMW uses a random forest model with a 95.3% F1 score to generate maps on a yearly cadence, covering only about half of relevant coastal regions. Using OlmoEarth Platform we fine-tune a OlmoEarth model using their data up to an F1 score of 98.1%. The OlmoEarth Platform can run inference on a monthly cadence to generate new maps, or on a rolling basis to detect change faster.

Global Ecosystem Atlas is building a comprehensive map of the world’s ecosystems [30]. For the last 3 months they have been using OlmoEarth Platform to label more than 15,000 data points. OlmoEarth Platform allows them

to partition areas of interest, generate points to label, assign those points to labelers, review the results, and export the data or fine-tune a model directly in the platform. With a subset of the data from North Africa we fine-tune a model that achieves state-of-the-art accuracy and run inference to generate new ecosystem maps (Figure 5). Humans can review the results to feed better labels back into the training pipeline.

5.2. Downstream Risks

The power and versatility of OlmoEarth also bring risks. We release OlmoEarth under an open license designed to address some of these risks by allowing the free use, modification, and sharing of the model weights, datasets, and associated code while restricting use for military, defense-related, and extractive industry applications.

5.3. The Future

We plan to add climate and weather data and forecasting to the OlmoEarth model to help with tasks like wildfire prediction and crop yield forecasting. Expanding to this kind of data will require handling a wider variety of input resolutions both spatially (from meters to kilometers) and temporally (from days to years).

We also plan to add non-geospatial data to the model. Often data labeling for tasks like crop type mapping requires in-person site visits and ground-level observation. Geolocated natural images would expand OlmoEarth’s ability to handle these fine-grained recognition tasks.

Ultimately we want to support and grow the community of partner organizations who bring incredible knowledge, expertise, and passion to this work. We plan to learn from our partners about what tools and capabilities they need and then improve OlmoEarth Platform to better help them. We hope OlmoEarth Platform can become a hub for data, models, training, and inference across a wide range of organizations working to solve the world’s biggest problems.

5.4. Conclusion

We present the OlmoEarth family of Earth observation foundation models, achieving state-of-the-art performance across 37 tasks compared to 12 existing models. Our training recipe, Latent MIM Lite, enables stable training in latent space through a frozen, randomly initialized target encoder, avoiding the representation collapse common in latent approaches. Combined with modality-aware masking and a modality-aware contrastive loss, this recipe produces strong representations from both observational and supervised map data. We deploy OlmoEarth as part of the OlmoEarth Platform, an open platform for data curation, model fine-tuning, and inference. All code, model weights, and training data are publicly available.

Acknowledgments

We wish to express deep gratitude to our early collaborators who shared data, expertise, and time to make these models successful for real-world, mission-critical applications: Amazon Conservation Association, African Wildlife Foundation, CGIAR/International Food Policy Research Institute (IFPRI), Global Mangrove Watch, Global Ecosystem Atlas, ITC University of Twente, NASA Jet Propulsion Laboratory (JPL), and NASA Harvest.

We would also like to thank the OLMo-core, Beaker, Comms, and Legal teams at Ai2 for their support, especially Pete Walsh, Dirk Groeneveld, Sam Skjonsberg, Tara Wilkins, Caroline Wu, Johann Dahm, David Albright, Kyle Wiggers, Jordan Steward, Crystal Nam, Will Smith, and Janice Dow.

References

- [1] Mahmoud Assran, Quentin Duval, Ishan Misra, Piotr Bojanowski, Pascal Vincent, Michael Rabbat, Yann LeCun, and Nicolas Ballas. Self-supervised learning from images with a joint-embedding predictive architecture. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 15619–15629, 2023. 2, 7
- [2] Guillaume Astruc, Nicolas Gonthier, Clement Mallet, and Loic Landrieu. AnySat: An Earth observation model for any resolutions, scales, and modalities. *arXiv preprint arXiv:2412.14123*, 2024. 1, 5, 7
- [3] Kumar Ayush, Burak Uzkent, Chenlin Meng, Kumar Tanmay, Marshall Burke, David Lobell, and Stefano Ermon. Geography-aware self-supervised learning. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 10181–10190, 2021. 7
- [4] Favyen Bastani, Piper Wolters, Ritwik Gupta, Joe Ferdinando, and Aniruddha Kembhavi. SatlasPretrain: A large-scale dataset for remote sensing image understanding. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 16772–16782, 2023. 5
- [5] Lucas Beyer, Pavel Izmailov, Alexander Kolesnikov, Mathilde Caron, Simon Kornblith, Xiaohua Zhai, Matthias Minderer, Michael Tschannen, Ibrahim Alabdulmohsin, and Filip Pavetic. FlexiViT: One model for all patch sizes. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 14496–14506, 2023. 2
- [6] Ella Bingham and Heikki Mannila. Random projection in dimensionality reduction: applications to image and text data. In *Proceedings of the seventh ACM SIGKDD international conference on Knowledge discovery and data mining*, pages 245–250, 2001. 4
- [7] Avrim Blum. Random projection, margins, kernels, and feature-selection. In *International Statistical and Optimization Perspectives Workshop "Subspace, Latent Structure and Feature Selection"*, pages 52–68. Springer, 2005. 4
- [8] Derrick Bonafilia, Beth Tellman, Tyler Anderson, and Erica Issenberg. Sen1Floods11: A georeferenced dataset to train and test deep learning flood algorithms for Sentinel-1. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops*, pages 210–211, 2020. 5
- [9] Christopher F Brown, Michal R Kazmierski, Valerie J Pasquarella, William J Rucklidge, Masha Samsikova, Chenhui Zhang, Evan Shelhamer, Estefania Lahera, Olivia Wiles, Simon Ilyushchenko, et al. Alphaeart foundations: An embedding field model for accurate and efficient global mapping from sparse label data. *arXiv preprint arXiv:2507.22291*, 2025. 3, 8, 15
- [10] Ting Chen, Simon Kornblith, Mohammad Norouzi, and Geoffrey Hinton. A simple framework for contrastive learning of visual representations. In *International conference on machine learning*, pages 1597–1607. PMLR, 2020. 4
- [11] Clay. Clay Foundation Model - Clay Foundation Model. <https://clay-foundation.github.io/model/>, 2025. 5
- [12] Yezhen Cong, Samar Khanna, Chenlin Meng, Patrick Liu, Erik Rozi, Yutong He, Marshall Burke, David Lobell, and Stefano Ermon. Satmae: Pre-training transformers for temporal and multi-spectral satellite imagery. *Advances in Neural Information Processing Systems*, 35:197–211, 2022. 7
- [13] Isaac Corley et al. Revisiting pre-trained remote sensing model benchmarks: Resizing and normalization matters. In *CVPR Workshops*, 2024. 5
- [14] Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. Bert: Pre-training of deep bidirectional transformers for language understanding. *arXiv preprint arXiv:1810.04805*, 2018. 7
- [15] Zhengpeng Feng, Sadiq Jaffer, Jovana Knezevic, Silja Sormunen, Robin Young, Madeline Lisaius, Markus Immitzer, James Ball, Clement Atzberger, David A Coomes, et al. Tessera: Temporal embeddings of surface spectra for earth representation and analysis. *arXiv preprint arXiv:2506.20380*, 2025. 1, 5, 8
- [16] Anthony Fuller, Koreen Millard, and James Green. CROMA: Remote sensing representations with contrastive radar-optical masked autoencoders. In *Advances in Neural Information Processing Systems*, 2024. 1, 2, 5
- [17] Vivien Sainte Fare Garnot and Loic Landrieu. Panoptic segmentation of satellite image time series with convolutional temporal attention networks. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 4872–4881, 2021. 5, 13
- [18] Ross Girshick. Fast r-cnn. In *Proceedings of the IEEE international conference on computer vision*, pages 1440–1448, 2015. 4
- [19] Dirk Groeneveld, Iz Beltagy, Pete Walsh, Akshita Bhagia, Rodney Kinney, Oyvind Tafjord, Ananya Harsh Jha, Hamish Ivison, Ian Magnusson, Yizhong Wang, Shane Arora, David Atkinson, Russell Authur, Khyathi Raghavi Chandu, Arman Cohan, Jennifer Dumas, Yanai Elazar, Yuling Gu, Jack Hessel, Tushar Khot, William Merrill, Jacob Morrison, Niklas Muennighoff, Aakanksha Naik, Crystal Nam, Matthew E. Peters, Valentina Pyatkin, Abhilasha Ravichander, Dustin Schwenk, Saurabh Shah, Will Smith, Emma Strubell, Nishant Subramani, Mitchell Wortsman, Pradeep

- Dasigi, Nathan Lambert, Kyle Richardson, Luke Zettlemoyer, Jesse Dodge, Kyle Lo, Luca Soldaini, Noah A. Smith, and Hannaneh Hajishirzi. Olmo: Accelerating the science of language models, 2024. 7
- [20] Group on Earth Observations (GEO). Global Ecosystems Atlas. <https://globalecosystemsatlas.org>, 2025. 14
- [21] Xin Guo, Jiangwei Lao, Bo Dang, Yingying Zhang, Lei Yu, Lixiang Ru, Liheng Zhong, Ziyuan Huang, Kang Wu, Dingxiang Hu, et al. Skysense: A multi-modal remote sensing foundation model towards universal interpretation for earth observation imagery. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 27672–27683, 2024. 8
- [22] Matthew Gwilliam and Abhinav Shrivastava. Beyond supervised vs. unsupervised: Representative benchmarking and analysis of image representation learning. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 9642–9652, 2022. 5
- [23] Kaiming He, Xinlei Chen, Saining Xie, Yanghao Li, Piotr Dollár, and Ross Girshick. Masked autoencoders are scalable vision learners. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 16000–16009, 2022. 7
- [24] Patrick Helber, Benjamin Bischke, Andreas Dengel, and Damian Borth. Eurosat: A novel dataset and deep learning benchmark for land use and land cover classification. *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing*, 12(7):2217–2226, 2019. 13
- [25] Hans Hersbach, Bill Bell, Paul Berrisford, Shoji Hirahara, András Horányi, Joaquín Muñoz-Sabater, Julien Nicolas, Carole Peubey, Raluca Radu, Dinand Schepers, et al. The era5 global reanalysis. *Quarterly Journal of the Royal Meteorological Society*, 146(730):1999–2049, 2020. 2
- [26] Jeremy Irvin, Hao Sheng, Neel Ramachandran, Sonja Johnson-Yu, Sharon Zhou, Kyle Story, Rose Rustowicz, Cooper Elsworth, Kemen Austin, and Andrew Y Ng. Forestnet: Classifying drivers of deforestation in indonesia using deep learning on satellite imagery. *arXiv preprint arXiv:2011.05479*, 2020. 13
- [27] Johannes Jakubik, Felix Yang, Benedikt Blumenstiel, Erik Scheurer, Rocco Sedona, Stefano Maurogiovanni, Jente Bosmans, Nikolaos Dionelis, Valerio Marsocci, Niklas Kopp, Rahul Ramachandran, Paolo Fraccaro, Thomas Brunschweiler, Gabriele Cavallaro, Juan Bernabe-Moreno, and Nicolas Longépé. TerraMind: Large-scale generative multimodality for earth observation. In *ICCV*, 2025. 1, 4, 5
- [28] Neal Jean, Sherrie Wang, Anshul Samar, George Azzari, David Lobell, and Stefano Ermon. Tile2vec: Unsupervised representation learning for spatially distributed data. In *Proceedings of the AAAI Conference on Artificial Intelligence*, pages 3967–3974, 2019. 7
- [29] Z. Jin, C. Lin, C. Weigl, J. Obarowski, and D. Hale. Smallholder cashew plantations in benin version 1.0. <https://doi.org/10.34911/rdnt.hfv20i>, 2021. 13
- [30] David A. Keith, José R. Ferrer-Paris, Emily Nicholson, Melanie J. Bishop, Beth A. Polidoro, Eva Ramirez-Llodra, Mark G. Tozer, Jeanne L. Nel, Ralph Mac Nally, and Edward J. Gehr. A function-based typology for earth’s ecosystems. *Nature*, 610:513–518, 2022. 8
- [31] Katerina Kikaki, Ioannis Kakogeorgiou, Ibrahim Hoteit, and Konstantinos Karantzas. Detecting marine pollutants and sea surface features with deep learning in Sentinel-2 imagery. *ISPRS Journal of Photogrammetry and Remote Sensing*, 210:39–54, 2024. 5, 13
- [32] Alexandre Lacoste, Nils Lehmann, Pau Rodriguez, Evan Sherwin, Hannah Kerner, Björn Lütjens, Jeremy Irvin, David Dao, Hamed Alemohammad, Alexandre Drouin, et al. GEO-Bench: Toward foundation models for earth monitoring. In *Advances in Neural Information Processing Systems*, 2024. 5
- [33] Jihyeon Lee, Nina R Brooks, Fahim Tajwar, Marshall Burke, Stefano Ermon, David B Lobell, Debashish Biswas, and Stephen P Luby. Scalable deep learning to identify brick kilns and aid regulatory capacity. *Proceedings of the National Academy of Sciences*, 118(17):e2018863118, 2021. 13
- [34] Tsung-Yi Lin, Piotr Dollár, Ross Girshick, Kaiming He, Bharath Hariharan, and Serge Belongie. Feature pyramid networks for object detection. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pages 2117–2125, 2017. 6
- [35] Oscar Manas, Alexandre Lacoste, Xavier Giró-i Nieto, David Vazquez, and Pau Rodriguez. Seasonal contrast: Unsupervised pre-training from uncurated remote sensing data. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 9414–9423, 2021. 7
- [36] Shentong Mo and Shengbang Tong. Connecting joint-embedding predictive architecture with contrastive self-supervised learning. In *The Thirty-eighth Annual Conference on Neural Information Processing Systems*, 2024. 2, 7
- [37] Jacob Morrison, Clara Na, Jared Fernandez, Tim Dettmers, Emma Strubell, and Jesse Dodge. Holistically evaluating the environmental impact of creating language models, 2025. 7
- [38] National Aeronautics and Space Administration (NASA) Earthdata. Shuttle Radar Topography Mission. <https://e4ftl01.cr.usgs.gov/MEASURES/SRTMGL1.003/>, 2018. 2
- [39] Team OLMo, Pete Walsh, Luca Soldaini, Dirk Groeneveld, Kyle Lo, Shane Arora, Akshita Bhagia, Yuling Gu, Shengyi Huang, Matt Jordan, Nathan Lambert, Dustin Schwenk, Oyvind Tafjord, Taira Anderson, David Atkinson, Faeze Brahman, Christopher Clark, Pradeep Dasigi, Nouha Dziri, Allyson Ettinger, Michal Guerquin, David Heineman, Hamish Ivison, Pang Wei Koh, Jiacheng Liu, Saumya Malik, William Merrill, Lester James V. Miranda, Jacob Morrison, Tyler Murray, Crystal Nam, Jake Poznanski, Valentina Pyatkin, Aman Rangapur, Michael Schmitz, Sam Skjonsberg, David Wadden, Christopher Wilhelm, Michael Wilson, Luke Zettlemoyer, Ali Farhadi, Noah A. Smith, and Hannaneh Hajishirzi. 2 olmo 2 furious, 2025. 7
- [40] OpenStreetMap contributors. Planet dump retrieved from <https://planet.osm.org>. <https://www.openstreetmap.org>, 2017. 2
- [41] Colorado J Reed, Ritwik Gupta, Shufan Li, Sarah Brockman, Christopher Funk, Brian Clipp, Kurt Keutzer, Salvatore

- Candido, Matt Uyttendaele, and Trevor Darrell. Scale-MAE: A scale-aware masked autoencoder for multiscale geospatial representation learning. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 4088–4099, 2023. 2, 5, 7
- [42] Shaoqing Ren, Kaiming He, Ross Girshick, and Jian Sun. Faster r-cnn: Towards real-time object detection with region proposal networks. *IEEE transactions on pattern analysis and machine intelligence*, 39(6):1137–1149, 2016. 6
- [43] Olaf Ronneberger, Philipp Fischer, and Thomas Brox. U-net: Convolutional networks for biomedical image segmentation. In *International Conference on Medical image computing and computer-assisted intervention*, pages 234–241. Springer, 2015. 6
- [44] Marc Rußwurm, Sébastien Lefèvre, and Marco Körner. BreizhCrops: A satellite time series dataset for crop type identification. In *Proceedings of the International Conference on Machine Learning Time Series Workshop*, 2019. 5, 13
- [45] R Siddharth and Gnanasekaran Aghila. Randpro-a practical implementation of random projection-based feature extraction for high dimensional multivariate data analysis in R. *SoftwareX*, 12:100629, 2020. 4
- [46] Oriane Siméoni, Huy V. Vo, Maximilian Seitzer, Federico Baldassarre, Maxime Oquab, Cijo Jose, Vasil Khalidov, Marc Szafraniec, Seungeun Yi, Michaël Ramamonjisoa, Francisco Massa, Daniel Haziza, Luca Wehrstedt, Jianyuan Wang, Timothée Darcet, Théo Moutakanni, Leonel Sentana, Claire Roberts, Andrea Vedaldi, Jamie Tolan, John Brandt, Camille Couprie, Julien Mairal, Hervé Jégou, Patrick Labatut, and Piotr Bojanowski. DINOv3, 2025. 5, 7
- [47] Gencer Sumbul, Marcela Charfuelan, Begüm Demir, and Volker Markl. Bigearthnet: A large-scale benchmark archive for remote sensing image understanding. In *IGARSS 2019-2019 IEEE International Geoscience and Remote Sensing Symposium*, pages 5901–5904. IEEE, 2019. 13
- [48] Xian Sun, Peijin Wang, Wanxuan Lu, Zicong Zhu, Xiaonan Lu, Qibin He, Junxi Li, Xue Rong, Zhujun Yang, Hao Chang, Qinglin He, Guang Yang, Ruiping Wang, Jiwen Lu, and Kun Fu. Ringmo: A remote sensing foundation model with masked image modeling. *IEEE Transactions on Geoscience and Remote Sensing*, 61:1–22, 2023. 8
- [49] Daniela Szwarcman, Sujit Roy, Paolo Fraccaro, Thorsteinn Elí Gíslason, Benedikt Blumenstiel, Rinki Ghosal, Pedro Henrique de Oliveira, Joao Lucas de Sousa Almeida, Rocco Sedona, Yanghui Kang, et al. Prithvi-eo-2.0: A versatile multi-temporal foundation model for earth observation applications. *arXiv preprint arXiv:2412.02732*, 2024. 5
- [50] Jamie Tolan, Hung-I Yang, Benjamin Nosarzewski, Guillaume Couairon, Huy V Vo, John Brandt, Justine Spore, Sayantan Majumdar, Daniel Haziza, Janaki Vamaraju, et al. Very high resolution canopy height maps from rgb imagery using self-supervised vision transformer and convolutional decoder trained on aerial lidar. *Remote Sensing of Environment*, 300:113888, 2024. 2
- [51] Gabriel Tseng, Ivan Zvonkov, Catherine Lilian Nakalembe, and Hannah Kerner. CropHarvest: A global dataset for crop-type classification. In *Thirty-fifth Conference on Neural Information Processing Systems Datasets and Benchmarks Track (Round 2)*, 2021. 5, 13
- [52] Gabriel Tseng, Ruben Cartuyvels, Ivan Zvonkov, Mirali Purohit, David Rolnick, and Hannah Kerner. Lightweight, pre-trained transformers for remote sensing timeseries. *arXiv preprint arXiv:2304.14065*, 2023. 5, 7
- [53] Gabriel Tseng, Anthony Fuller, Marlena Reil, Henry Herzog, Patrick Beukema, Favyen Bastani, James R Green, Evan Shelhamer, Hannah Kerner, and David Rolnick. Galileo: Learning global & local features of many remote sensing modalities. In *Forty-second International Conference on Machine Learning*, 2025. 1, 2, 4, 5, 7, 16
- [54] United States Department of Agriculture (USDA) National Agricultural Statistics Service (NASS). Cropland Data Layer: USDA NASS, 2024. National Agricultural Statistics Service Marketing and Information Services Office, Washington, D.C. Retrieved from Link: <https://croplandcros.scinet.usda.gov/>. 2
- [55] U.S. Geological Survey. National agriculture imagery program: 2003 - present. <https://doi.org/10.5066/F7QN651G>, 2023. 2
- [56] Kristof Van Tricht, Jeroen Degerickx, Sven Gilliams, Daniele Zanaga, Marjorie Battude, Alex Grosu, Joost Brombacher, Myroslava Lesiv, Juan Carlos Laso Bayas, Santosh Karanam, et al. WorldCereal: a dynamic open-source system for global-scale, seasonal, and reproducible crop and irrigation mapping. *Earth System Science Data Discussions*, 2023:1–36, 2023. 2
- [57] Leonard Waldmann, Ando Shah, Yi Wang, Nils Lehmann, Adam Stewart, Zhitong Xiong, Xiao Xiang Zhu, Stefan Bauer, and John Chuang. Panopticon: Advancing any-sensor foundation models for earth observation. In *Proceedings of the Computer Vision and Pattern Recognition Conference (CVPR) Workshops*, 2025. 1, 5, 7, 8
- [58] Yi Wang, Zhitong Xiong, Chenying Liu, Adam J Stewart, Thomas Dujardin, Nikolaos Ioannis Bountos, Angelos Zavras, Franziska Gerken, Ioannis Papoutsis, Laura Leal-Taixé, et al. Towards a unified copernicus foundation model for earth vision. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2025. 1, 5
- [59] Yibing Wei, Abhinav Gupta, and Pedro Morgado. Towards latent masked image modeling for self-supervised visual representation learning. In *ECCV*, 2024. 2, 7
- [60] Western Cape Department of Agriculture. Crop type classification dataset for western cape, south africa. <https://staging.source.coop/radianteearth/south-africa-crops-competition>, 2021. 13
- [61] Charith Wickrema et al. Scaling remote sensing foundation models: Data domain tradeoffs at the peta-scale, 2026. 7
- [62] Zhitong Xiong, Yi Wang, Fahong Zhang, Adam J Stewart, Joëlle Hanna, Damian Borth, Ioannis Papoutsis, Bertrand Le Saux, Gustau Camps-Valls, and Xiao Xiang Zhu. Neural plasticity-inspired foundation model for observing the earth crossing modalities. *arXiv preprint arXiv:2403.15356*, 2024. 8
- [63] Marta Yebra, Gianluca Scortechini, Karine Adeline, Nursema Aktepe, Turkia Almoustafa, Avi Bar-Massada,

María Eugenia Beget, Matthias Boer, Ross Bradstock, Tegan Brown, et al. Globe-LFMC 2.0, an enhanced and updated dataset for live fuel moisture content research. *Scientific data*, 11(1):332, 2024. [14](#)

- [64] Daniele Zanaga, Ruben Van De Kerchove, Dirk Daems, Wanda De Keersmaecker, Carsten Brockmann, Grit Kirches, Jan Wevers, Oliver Cartus, Maurizio Santoro, Steffen Fritz, et al. ESA WorldCover 10 m 2021 v200. *ESA WorldCover Project*, 2022. [2](#)
- [65] Xiao Xiang Zhu, Jingliang Hu, Chunping Qiu, Yilei Shi, Jian Kang, Lichao Mou, Hossein Bagheri, Matthias Haberle, Yuansheng Hua, Rong Huang, et al. So2sat lc42: A benchmark data set for the classification of global local climate zones [software and data sets]. *IEEE Geoscience and Remote Sensing Magazine*, 8(3):76–89, 2020. [13](#)